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ORIGINAL PAPER

# DInSAR–SBAS satellite monitoring of infrastructures: how temperature affects the "Ponte della Musica" case study

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**Abstract**
Continuous monitoring of the structural health of strategic structures and transport infrastructures plays a crucial role in providing an effective assessment of the safety conditions and in timely planning of the ordinary and extraordinary maintenance programme. Deformation monitoring and dynamic characteristic identification are some commonly used strategies for this purpose. One of the main challenges of recent years in the field of structural health monitoring is the use of data deriving from satellite interferometry, capable of providing information on structural deformations at a local and territorial scale. Despite the solidity and dependability of satellite-based methods for assessing ground deformation over time, when it comes to structural surveillance, there are certain circumstances under which satellites are incapable of accurately assessing displacements. This is particularly true for structures that are sensitive to temperature variations. The paper uses the "Ponte della Musica–Armando Trovajoli" in Rome as a case study to explore these aspects in more depth. This bridge has a steel arch structure with a prestressed concrete deck below it. It represents an example in which satellite differential interferometry does not allow obtaining useful information on displacement, at least for the most deformable portion of the deck, and therefore also on any pathological movements. This work proposes a 3D digital twin of the bridge, appropriately calibrated through experimental measurements of the environmental vibrations performed on the bridge. This will allow to evaluate the role played by thermal deformations related to air temperature variations and thus better understand the connection between physiological deformations and satellite limits.

**Keywords** Structural health monitoring · Satellite interferometry · In situ measurements · Numerical analysis · Infrastructural thermal deformations

## 1 Introduction

The management and maintenance of transport infrastructures is a key issue in European regions, particularly for twentieth century reinforced concrete structures. Damage accumulation and resistance reduction are associated with various factors such as ageing and material deterioration, traffic cyclic load, environmental excitation, earthquakes, and landslides. To tackle this problem organically, a proposal for procedures and tools for assessing the safety and maintenance of transport infrastructures have recently been summarized in the new guidelines issued by the Ministry of Infrastructure and Transport in 2020 [1].

Identification of structural modal parameters, including natural frequencies, modal damping, and mode shapes, is currently one of the most effective strategies for bridge safety monitoring. Significant progress has been made in the field of structural health monitoring and dynamic identification, based on different approaches and techniques. Several authors [2–4] have analysed various engineering algorithms and applications of data acquisition, data diagnosis and reconstruction of the structural health. Different studies were performed on dynamic identification based on the analysis of the variations of wave propagation, as reported by [5–7], and on the techniques operating in the time–frequency domain, as reported by [8–10]. It is well known that the presence of structural damage can change the dynamic characteristics of the structural system. Several methods for damage detection and localization on framed structures, based on the evaluation of the variation of modal or non-modal parameters and

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on the evaluation of the modal curvature evolution over time, were developed, such as by [10–17].

It has recently been observed that adequate integration of in situ structural health monitoring (SHM) data with interferometric satellite data could facilitate the management of critical infrastructures, the conservation of the built environment and the planning of interventions for risk mitigation [18, 19]. Remote sensing can provide valuable information on natural and anthropic hazard scenarios thanks to its synoptic capability. Numerous scientific studies have demonstrated the advantages of satellites equipped with synthetic aperture radar (SAR) sensors capable of monitoring a wide range of territorial-scale processes, such as subsidence, foundation failure and deformation of buildings and bridges [20]. SAR monitoring has the great advantage of working continuously for the time scale of the low-frequency deformation, with the ability to provide high-resolution weather-independent and light-independent imagery of the Earth (using an average sampling frequency equal to two weeks). Recent years have seen the development of new SAR sensors, characterized by technological advancements both in terms of spatial resolution, which today reaches the meter scale, and of a shorter re-entry period, of the order of a few days, aiming to obtain reliable data on structural safety conditions. The enhanced imaging capabilities of these new SAR sensors result in an impressive increase in the density of monitored targets, both in urban and rural areas [21]. Differential synthetic aperture radar (DInSAR) satellite interferometry is one of the most used and consolidated remote sensing techniques for detecting ground movements. It is also becoming one of the most innovative methodologies for monitoring the movements of structures in urban areas [22]. The technique is based on the utilization of the phase difference, also known as an interferogram, between two SAR images that are separated by time. Using the phase unwrapping operation, it allows recovering information on the displacements projected along the line of sight (LOS) of the radar sensor, which occurred between the two acquisition times. Millimetric-level motion on ground structures can be detected and measured with interferometric SAR technique [23, 24]. It is possible to use the deformations monitored through DInSAR techniques, which are typically slow and mainly related to ground deformations, for a preliminary structural assessment of the built environment. This may also lead to the development of integrated systems for early warning, monitoring, and rapid damage assessment of the built environment and critical infrastructure. Several examples of this integrated approach have been applied, focusing on its suitability for long-term monitoring of transport infrastructure [25–27].

However, it is worth noting that in the case of temporally or spatially fast varying deformations, the DInSAR technique may fail in correctly detecting the overall magnitude of displacements. This leads to an underestimation of the deformation measurements or even the absence of data [28]. Recent studies demonstrated that some improvement of the methodology for the assessment of single structure deformations phenomena could be obtained by combining COSMO-SkyMed (CSK) and Sentinel (SNT) InSAR measurements to evaluate millimetre scale deformation processes [28]. A numerical and experimental campaign was conducted as part of the ReLUIS 2019–2021 and 2022–2024 projects (WP6 structural health monitoring and satellite data) developed under the Italian Civil Protection Department agreement. One of the objectives of the research was to ascertain any limitations of the satellite evaluation of the movements of structures, in particular those that are very deformable or sensitive to temperature variations. The selected case study was the "Ponte della Musica Armando Trovajoli" bridge located in Rome, described by [29, 30]. This case study was interesting for engineering because satellite DInSAR measurements could not capture the central span of the bridge, even though steel has a high electromagnetic backscatter. The measurements only covered the end sections close to the side spans.

The study was divided into three main phases. In the initial observation phase, DInSAR remote sensing measurements provided by the Italian Space Agency (ASI) and processed by the CNR-IREA using the SBAS technique were used [31, 32]. The data obtained concerned the long-term deformation state of both the analysed structure and the surrounding ground. The analysis of the satellite data, relating to both the ascending and descending datasets, has instead highlighted the absence of satellite measurements on the central part of the deck, as reported by [29]. The same analyses, carried out on more rigid masonry arched bridges located near the "Ponte della Musica", report data both on the support and along the deck.

In a second phase, two experimental campaigns, integrated with a visual survey of the bridge, were carried out in two separate periods (October 2020 and November 2021) to measure ambient dynamic vibrations. The analyses of the velocimetric data acquired during the initial campaign, as explicated in [33], and the accelerometric data obtained during the second campaign have yielded valuable insights into the primary dynamic characteristics of the bridge. These include natural frequencies and equivalent viscous damping.

The third phase used these dynamic parameters to accurately calibrate a 3D digital twin of the bridge. This was then used to estimate the trend of structural deformations due to external loads, such as the air temperature variations, to try to correlate them with the lack of coherence between the DInSAR measurements observed experimentally.

## 2 The "Ponte della Musica-Armando Trovajoli" case study

The bridge named "Ponte della Musica—Armando Trovajoli" is an arch bridge designed for pedestrians and cyclists, which spans the Tiber River in Rome with a single span (Fig. 1). The bridge is approximately 187 m long, with a main span of 127 m and two side spans of 30 m (Fig. 2). Its width varies from 17.20 m at the abutments to 20.70 m at the centreline section. Both arches of the bridge are inclined outward and are made of tubular steel with an asymmetric drop section. The arches are connected to the deck through *U*-shaped connections formed by steel pendants that support rigid transverse structures placed at 8.5-m intervals. With a span-to-rise ratio of about 12:1 [33], the arches rise 10.6 m above the main bridge span. Construction began in 2008 and was completed in May 2011.

The deck beams comprise two pairs of girders, with the main beams exhibiting dissymmetrical z-shapes that are mirrored regarding the longitudinal axis (Fig. 3a) and the edge beams exhibiting closed box sections (Fig. 3b). The four continuous longitudinal beams perform the arch bridge tie rod. On the edges of the deck, steel curbs with box sections have also been built. The tubular section of the arch consists of a shape called "asymmetrical drop", 1837.5 mm wide, made of S355 steel grade (Fig. 3c). The



**Fig. 1** Structural details of "Ponte della Musica Armando Trovajoli" in Rome

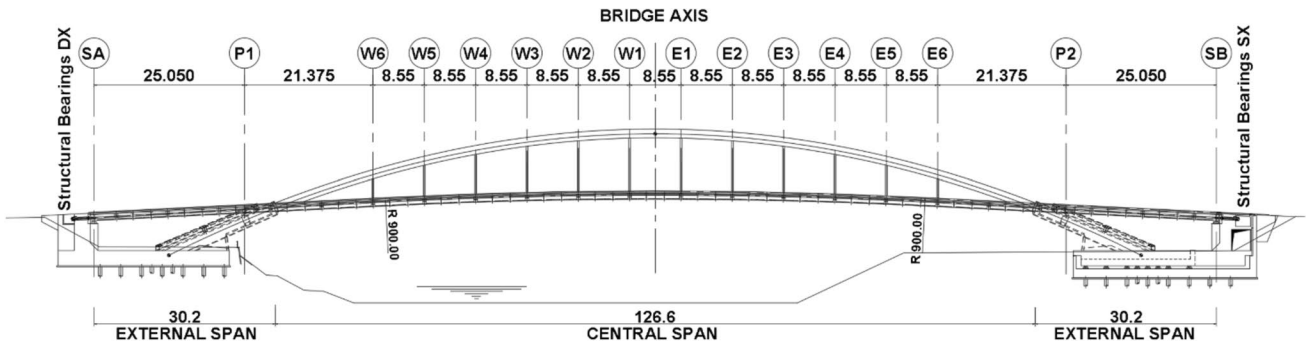



**Fig. 2** Structural scheme of the "Ponte della Musica" (design documentation provided by Ing. Stroveglia)

section of arches is also rotated by 21.2496°, to be orthogonal to the axis of the hangers, which are also rotated by the same angle regarding the vertical. The configuration of the arches appears curved, but it is a series of short straight elements, 4.5 m long, welded together.

The cross girders are subdivided into main cross girder, in correspondence with the steel tendon, forming the U-shaped rigid connection that links the arches with the deck. Secondary infill crossbeams are present in the central and outer areas of the bridge (Fig. 4). The crossbeams are shaped with a double flange cross section that varies in height along the cross-sectional dimension of the deck. The maximum height is between the edge beam and the central beam and is equal to the height of the latter, i.e. 1300 mm, then shortening towards the outside, where the minimum height is reached (250 mm) near the curb (Fig. 5).

A deck zone 6.6 m width was made of a high strength Rck 70/85 concrete mix of 180 mm height, allowing a reduction of the slab height and of the permanent gravity load. Flexible asphalt flooring was created on the reinforced concrete slab, designed for the purposes of a possible change of intended use from pedestrian traffic to road/railway traffic. In the external areas, we find flooring with non-slip Bankirai wooden slats for pedestrians only. Ponte della Musica bridge spans the Tiber River in the middle of a bend that shapes the northern part of the river's urban course [30]. The bridge is in the alluvial plain of the river. The substructure consists of foundations on piles due to the local geological and geotechnical setting of the subsoil (Fig. 6). In particular, the foundation consists of 54 piles (F 600) inserted in two concrete footing, 1620 $m^3$ on the right and 2785 $m^3$ on the left. As described in [30], during the development of the project, a fundamental change was defined in the structural

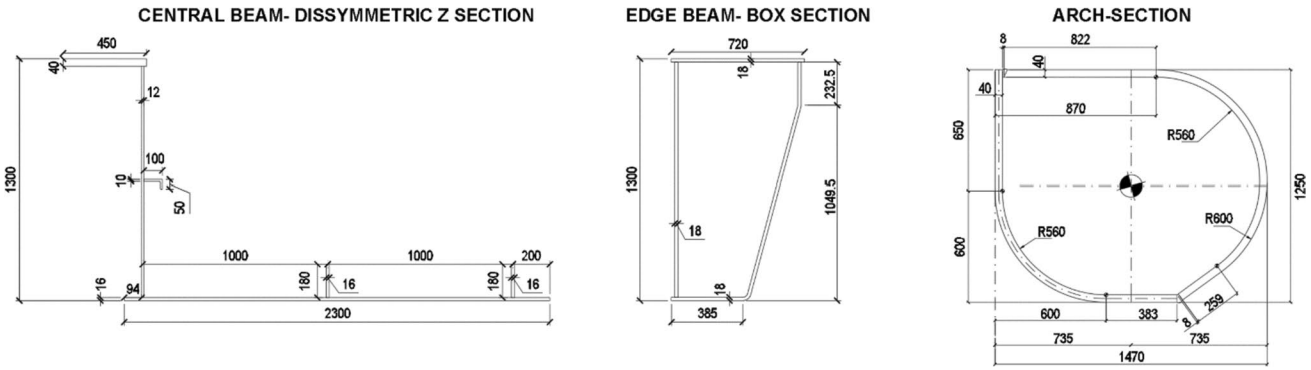



**Fig. 3** Sections of central beams; edge beams; arch section (design documentation provided by Ing. Stroveglia)

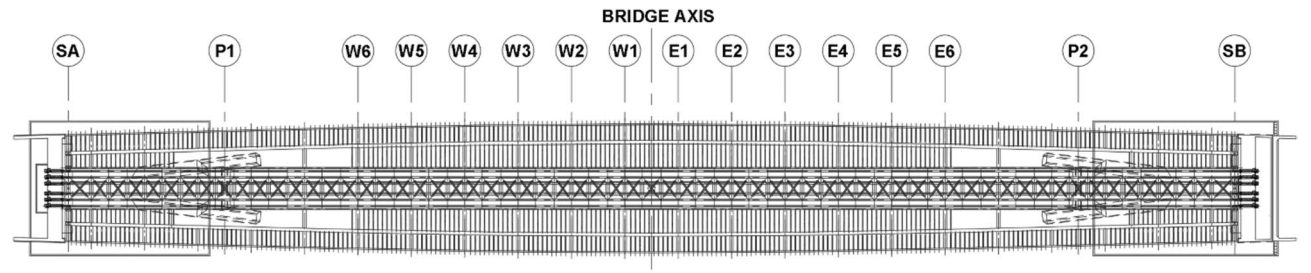



**Fig. 4** Plan view of the bridge deck (design documentation provided by Ing. Stroveglia)

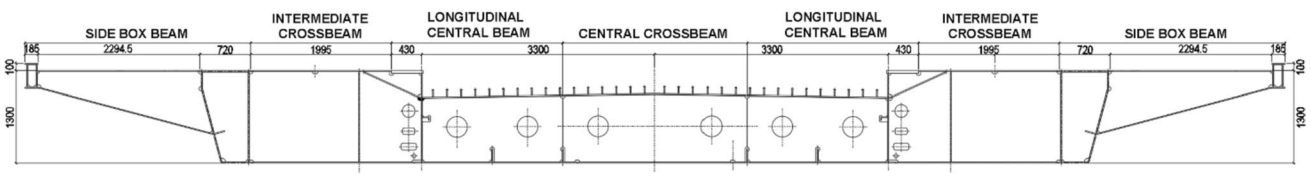



**Fig. 5** Transversal Type section of the bridge (design documentation provided by Ing. Stroveglia)

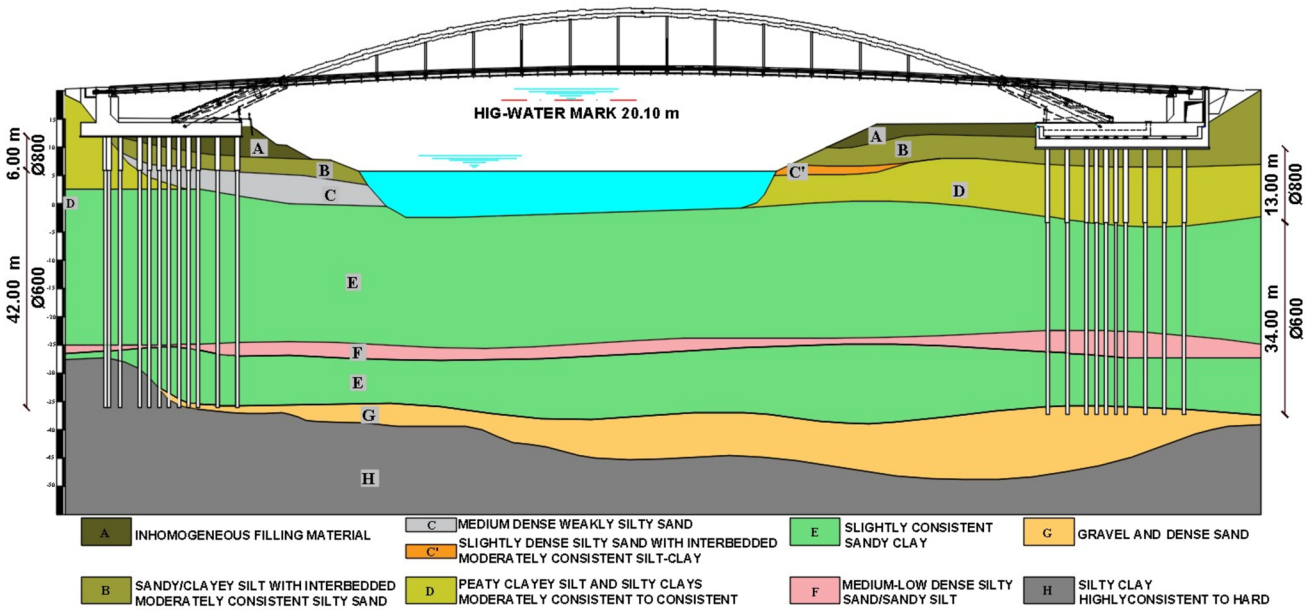



**Fig. 6** Geological section of the site [29]

scheme of the bridge, from a real arch to a tied arch, to avoid the solution of inclined mini piles (Fig. 6).

## 3 DInSAR for structural monitoring of the case study

The selected case study named "Ponte della Musica" is particularly interesting due to the absence of DInSAR satellite data on the central span, despite the high electromagnetic backscatter characteristic of steel. Furthermore, the bridge is not subject to vehicular traffic, thereby enabling the precise identification of the principal stresses and deformations because of the variations in the thermal loads due to air temperature variations. Full-resolution DInSAR multi-temporal technique [3] was used to monitor the investigated bridge. Specifically, ascending, and descending datasets provided by the Italian COSMO-SkyMed (CSK) satellite constellation were used. The algorithm considered was based on the small baseline subset (SBAS) approach [31, 32], generating mean deformation velocity maps and displacement time series from the SAR data, as reported in [29]. Originally, the SBAS algorithm aimed to investigate deformation phenomena on a territorial scale, to generate maps of a large area. It was then modified to investigate phenomena on a local scale to generate maps with full spatial resolution of a few metres, suitable for monitoring the time evolution of surface movements of urban areas, using an average sampling frequency of 2 weeks. The time coherence parameter is a key parameter, as it provides an estimate of the displacement for each individual measured point. This is done by determining the quality of the time series and evaluating the similarity between the initial strain signals and the chosen phase model. The time coherence value is expressed on a scale from 1 to 0, where the value 1 expresses the maximum similarity and the value 0 the zero similarity. Pixels characterized by a coherence value lower than a certain threshold, established according to specific interferometric analysis criteria, are considered unreliable and are therefore discarded in SBAS-DInSAR processing at full spatial resolution [32]. The acquisition mode used for the radar images was the standard Stripmap, particularly suitable for urban structures and individual DInSAR analysis. The characteristics of the technique are a spatial resolution on the ground of about 3 m both in azimuth (direction of the trajectory) and in the orthogonal direction. The two datasets are composed of 129 ascending and 107 descending single look complex (SLC) acquisitions, collected along the time interval March 2011–March 2019, with a viewing angle at the centre of the scene of about 34° and 29°, respectively. The available data, provided by ASI and preliminarily elaborated by CNR-IREA, were analysed in GIS software and classified according to average strain rate values. Figure 7 shows the average velocity maps, provided by CNR-IREA, measured on the infrastructure considering the ascending and descending datasets [29]. The deformation rate varies from – 0.4 to 0 cm/year and, as highlighted in Fig. 7, the points detected by the satellites are only on the external spans of the bridge but not on the central deck.

To investigate the reason for the absence of persistent scatterers (PS), the observation scale was expanded from a local scale (few hundreds of square metres) to a territorial scale (about of 1200 ha) in the northern area of Rome. The

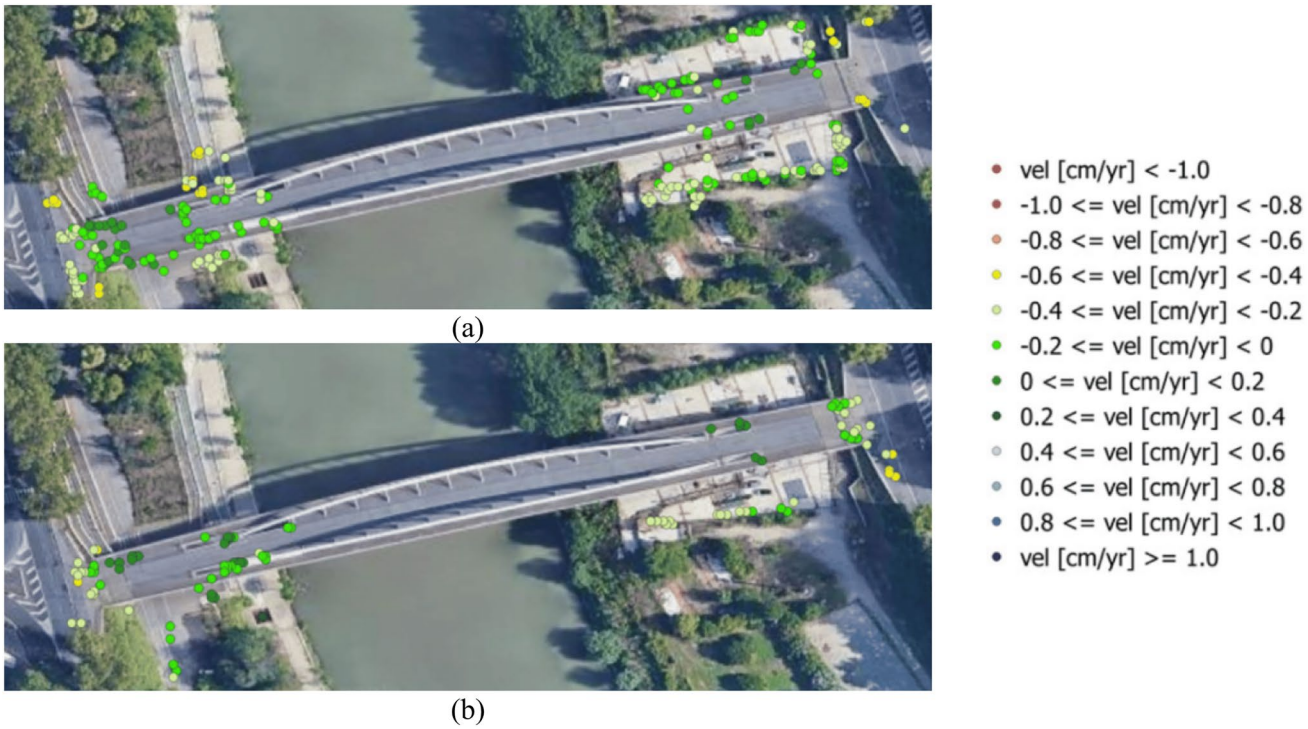



**Fig. 7** Deformation rate map of the "Ponte della Musica": **a** ascending orbits, **b** descending orbits [29]

same circumstance was observed for the other two bridges located in the area, such as Risorgimento bridge and Duca d'Aosta bridge, while satellite data was found on other structures, such as the Matteotti bridge, Pietro Nenni bridge, Flaminio bridge and Milvio bridge [29]. In general, it is possible that the reason for the absence of PS could be due to a combination of several factors. These include the high thermal deformability of bridges and the wavelength of the signal used (X band) for the satellite monitoring, as well as other factors, such as the shape and the limited reflectivity of the deck.

Starting from the differential interferogram, assuming that we have no topographic phase residues and that we can neglect, under appropriate considerations, the phase contributions due to the atmosphere, orbital errors and noise, the measured phase difference $\Delta\varphi$ consists only of $\Delta\varphi^{\text{displacement}}$ contribution. It is linked to the variation in the distance between sensor and target associated with the displacement of the points in the time interval between the two images, measured along the LOS of the sensor (represented by $d_{LOS}$), and can be reasonably expressed as:

$$\Delta\varphi = \Delta\varphi^{\text{displacement}} \approx \frac{4\pi}{\lambda} \cdot d_{\text{LOS}}, \tag{1}$$

where $\lambda$ it is the length of the electromagnetic wave emitted by the satellite linked to the satellite sensor used. By inverting the relationship (1) it is possible to obtain the measurement of the movement that occurred along the LOS $d_{\text{LOS}}$:

$$d_{\text{LOS}} \approx \frac{\Delta\varphi}{2\pi} \cdot \frac{\lambda}{2}. \tag{2}$$

From Eq. (2) it follows that a phase variation of $2\pi$ (i.e. an entire interferometric fringe) corresponds to a shift along the LOS of $\lambda/2$. Furthermore, it is observed that the displacement that we can measure appears to be directly connected to the interferometric phase, of which, however, only one main determination is originally known (the phase values belong to the interval $[-\pi, \pi]$). To recover the absolute phase variation from the pairs of SAR images, an operation called phase unwrapping is needed. This operation converts the interferometric phase from the "wrapped" phase to the "unwrapped" phase. The wrapped phase is the phase that is limited to the range of $-\pi$ to $+\pi$ radians, while the

**Table 1** Some of frequency range used by satellite radar systems

| Designation | Frequency range (GHz) | Wavelength range (m) |
|---|---|---|
| Banda L | 0.390–1.550 | 0.7690–0.1930 |
| Banda C | 3.900–5.750 | 0.0769–0.0520 |
| Banda X | **5.75–10.90** | **0.0520–0.0275** |

Bold represents the frequency band used for satellite acquisitions

unwrapped phase is the phase that can have any value. The surface displacements can be measured with an accuracy of a fraction of the wavelength of the SAR sensor. Table 1 provides examples of the typical frequency range used by sensors of satellites for interferometric analyses.

## 4 In situ investigations

As described in previous paragraphs, with the aim of investigating the reasons related to the absence of satellite data in the main span of the monitored bridge, experimental observations must be supported by numerical investigations with the aim of evaluating the structural displacement produced by the applied loads. To carry out consistent evaluations, the numerical model (described in the Sect. 5) was appropriately calibrated using real modal parameters recovered experimentally on the bridge. For this, as described in Sect. 4.1, vibrational-based investigations have been used to evaluate experimental modal parameters. Rather, the data used for thermal numerical analyses has been determined through the information obtained from a weather meteorological station (named Boncompagni) situated at 3600 m from the bridge.

### 4.1 Vibrational monitoring of the bridge and related experimental results

The bridge's structural dynamic characteristics were investigated using two experimental approaches. During the first experimental campaign, a single mobile velocimetric station has been used, as described in [29] and here not discussed. The second experimental campaign, which is more accurate, employed multiple three-directional accelerometric stations. Activities were performed in November 2021 [33, 34] and involved two sensor configurations, each consisting of six synchronized three-directional accelerometric stations positioned along the bridge in two distinct geometric arrangements, as depicted in Fig. 8a, b.

Within the accelerometric experimental campaign, three 1-h duration tests were performed using a 250 Hz sampling frequency. During the initial two tests, the sensor configuration depicted in Fig. 8a was employed, whereas for the third test, the configuration depicted in Fig. 8b was employed. Force-balanced accelerometric sensors with a 24-bit digitizer and a wide dynamic range were used. As depicted in Fig. 9, the maximum acceleration measured along the vertical component is superior to that measured along the longitudinal and transversal components.

Analysing the accelerometric time histories recorded on the investigated bridge by using the frequency domain decomposition (FDD) technique [35], the first three structural eigenfrequencies and related mode shapes were evaluated, while the equivalent viscous damping factors associated with the identified eigenmodes were evaluated using the non-parametric damping analysis [36]. Table 2 summarizes the main experimental dynamic characteristics of the bridge.

Figure 10 shows the first three identified mode shapes of the investigated bridge.

Experimental dynamic parameters (eigenfrequencies, eigenmodes and equivalent viscous damping factors) retrieved from the second experimental campaign were used for calibrating the numerical model.

### 4.2 Analysis of temperature data of Boncompagni weather station

Available climate data were used to conduct a simplified assessment of the impact of varying environmental temperatures on the deformation state of the bridge. The temperature data [37] refer to a period from January 2013 to December 2021 and were recorded at the meteorological station 4 (AL007 – Boncompagni), which is the closest to the bridge location, at 3600 m (Fig. 11). The Boncompagni station measures meteorological parameters with a time step of 30 min, resulting in 48 measurements every 24 h.

As depicted in Fig. 12, it is evident that the temperature variations exhibit a periodic trend, corresponding to




**Fig. 8** Accelerometer sensor layout of the second ambient vibration recordings (November 2021): **a** sensor configuration n.1; **b** sensor configuration n.2

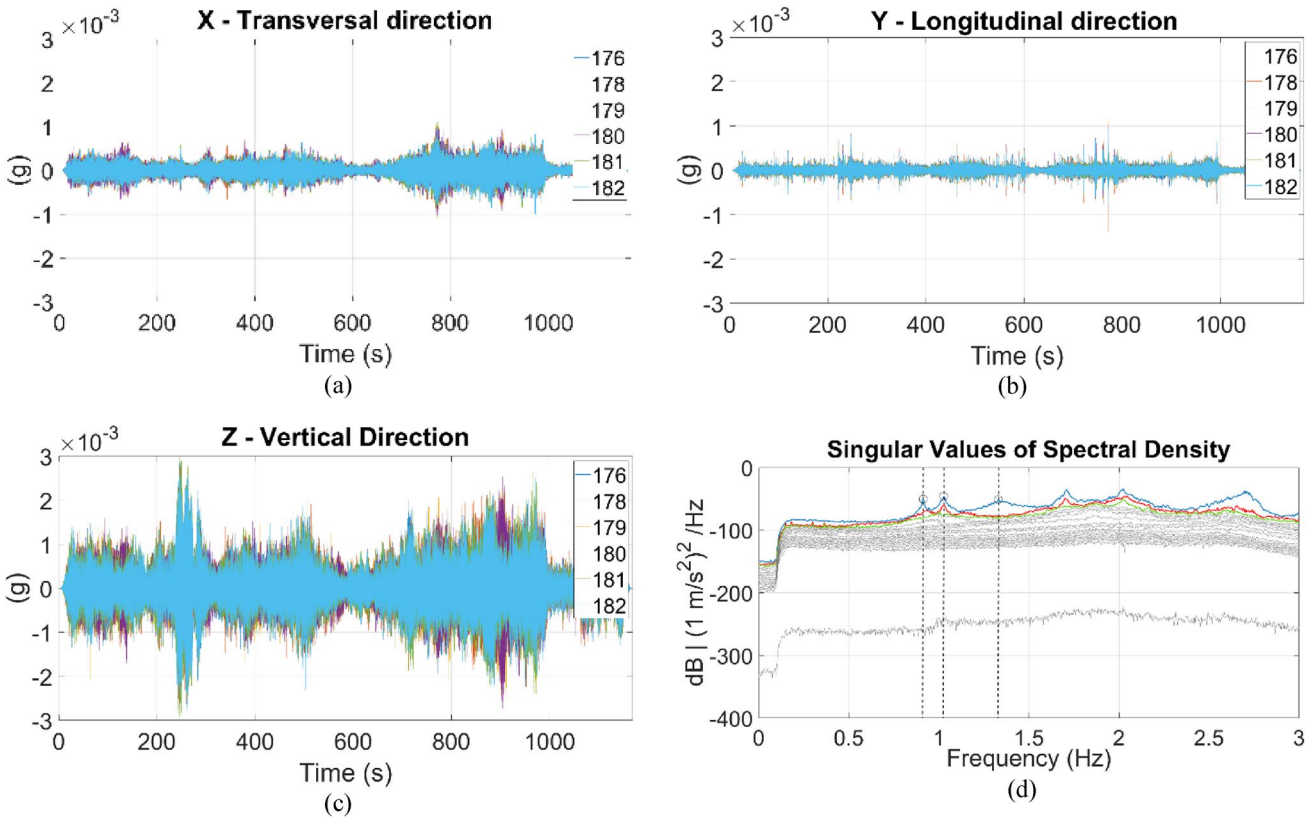



**Fig. 9** Accelerometric recordings (20 min) related to the second sensors' configuration during the 2021 experimental campaign: **a** transversal direction, **b** longitudinal direction, **c** vertical direction; **d** detection of the first three eigenfrequencies from the singular values of spectral density

**Table 2** Main experimental eigenfrequencies and equivalent viscous damping factors of the investigated bridge

| Mode number | Frequency [Hz] | Equivalent viscous damping factor [%] | Mode characteristics |
|---|---|---|---|
| Mode 1 | $0.92 \pm 0.01$ | 4 | Translational (vertical) |
| Mode 2 | $1.01 \pm 0.01$ | 4 | Rotational (longitudinal axis) |
| Mode 3 | $1.33 \pm 0.01$ | 4 | Translational (vertical) |

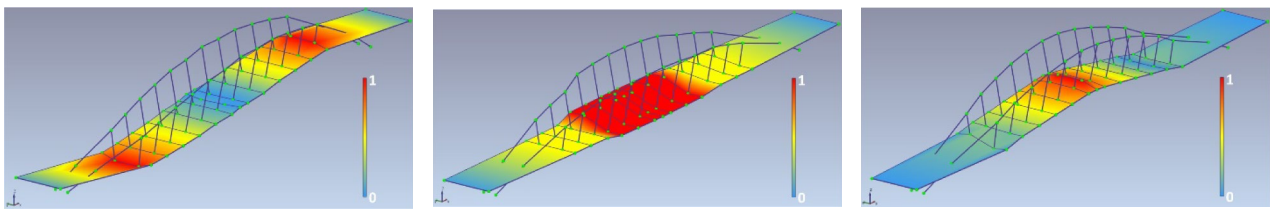



**Fig. 10** Experimental mode shapes of the investigated bridge: **a** fundamental mode, **b** second mode and **c** third mode

seasonal and sunlight variations, with the lowest values recorded in January and the highest values recorded in July and August. Instead, by utilizing the frequency representation of the recorded temperature, it is feasible to quantify the recurrence of the primary temperature variations. The fundamental peak of the Fourier transform is associated with the annual variations, but it is also significant to consider the daily temperature variations.

To precisely determine the temperature difference to be used in the numerical analyses, the real temperature




**Fig. 11** Geographical location of the reference weather station 4 (AL007–Boncompagni)— from Google Earth

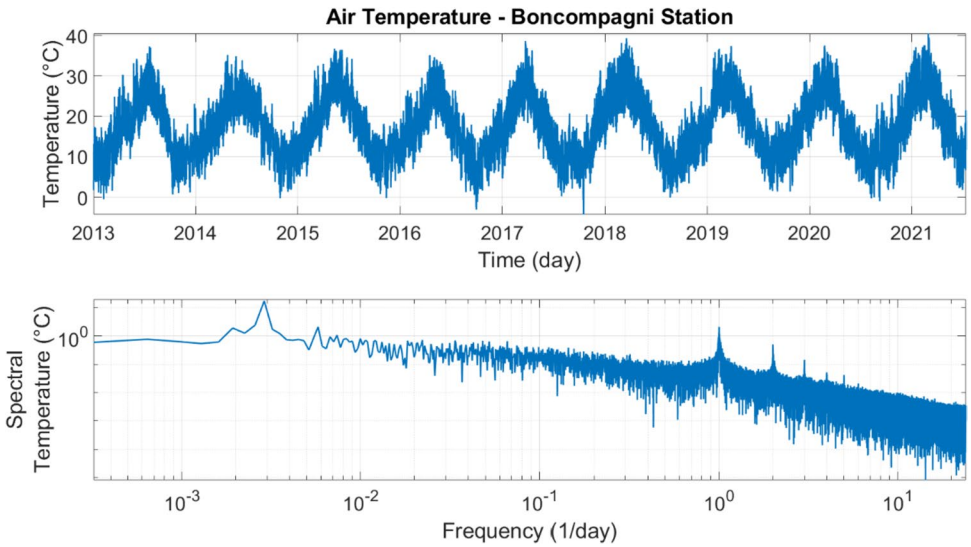



**Fig. 12** Representation of the temperature history over 8 years (2013–2021) both in time (top) and frequency domain (bottom)

variations in relation to the satellite's acquisitions along the ascending and descending orbits were considered. Satellite acquisitions occur on average every 16 days, and the acquisition time is fixed at 5.45 a.m. for the ascending orbit and 6.09 p.m. for the descending orbit. Figure 13 illustrates the history of the temperatures detected in correspondence with the satellite acquisitions and the related spectral variations. It is observable that the modulus of the average temperature difference between the two orbits amounts to approximately 5 °C, based on a statistical analysis of the time histories of the temperatures analysed in accordance with the satellite passes. Furthermore, this disparity can be swiftly discerned through a meticulous examination of the temperature histories depicted in Fig. 13. The temperature values of the descending orbit (represented by the black line) are consistently higher than those of the ascending orbit (represented by the red line).

## 5 Numerical modelling

Tto perform thermo-mechanical analyses of the "Ponte della Musica—Armando Trovajoli", a 3D finite element numerical model was implemented in Sap2000 [38], as shown in Fig. 14. The FE model has been built using frame elements for beams, steel tendons, curbs, arche, and shell elements for the slab, support raft and shoulders. Prestressed cables were modelled as a tendon element using an equivalent section made of harmonic steel. The element geometries, construction details, material characteristics, loads and section

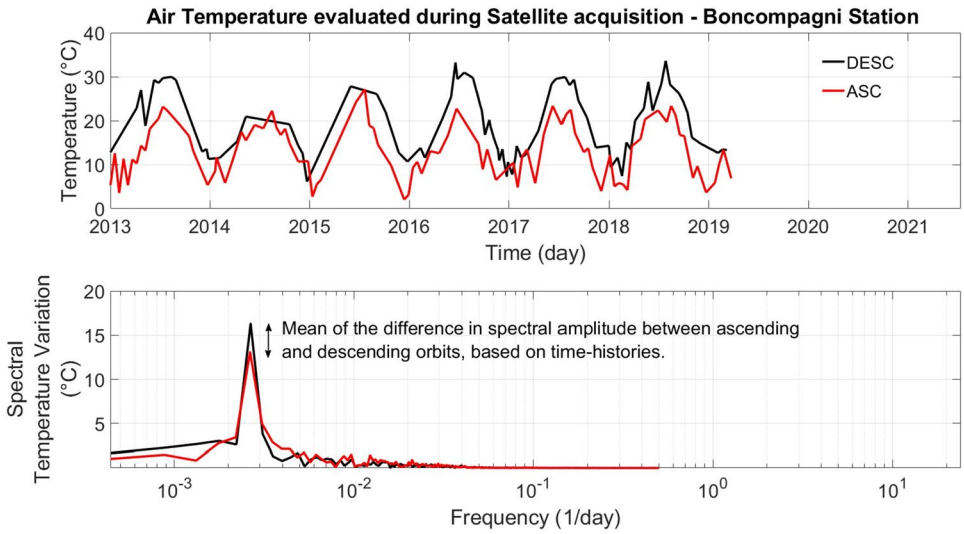



**Fig. 13** Representation of the ambient temperature measured in correspondence to the satellite acquisitions along ascending and descending orbits in both time (top) and frequency domain (bottom)

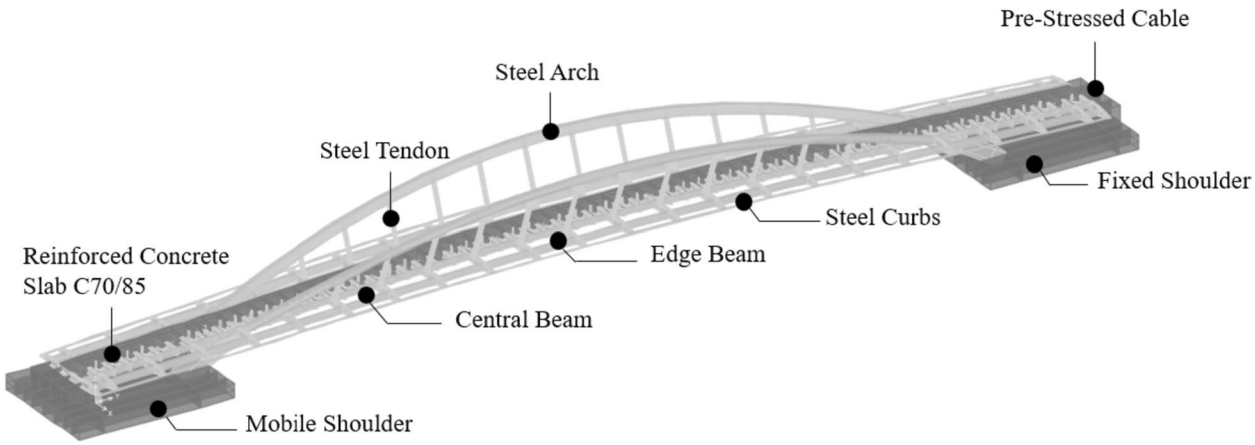



**Fig. 14** 3D FEM model of the "Ponte della Musica"

typologies of the structural elements were determined based on the available design documentation.

The model calibration used data from the original design documentation for permanent loads and constraints (verified during visual inspections). According to the load conditions observed during the experimental campaign, accidental overloads were not considered. For the deck, link elements were used within the numerical model to simulate the boundary conditions representative of the real situations detected during ambient vibration measurements. For arches and abutments, fixed constraints were used. Table 3 presents the element materials of the bridge considered for the numerical modelling.

Several constraint and boundary conditions have been considered for a proper calibration of the FEM model of the bridge, as described in [33, 34]. In these papers, outcomes concerning the ambient vibration conditions (microtremors)

**Table 3** Materials used within the numerical model

| Element | Material |
|---|---|
| Arch | S355 steel |
| Tendon | S355 steel |
| Edge beam | S355 steel |
| Central beam | S355 steel |
| Crossbeam | S355 steel |
| Curbs | S355 steel |
| Horizontal bracing | S355 steel |
| Slab | C70/85 reinforced concrete |
| Prestressed cables | Harmonic steel |
| Mobile shoulder | C70/85 reinforced concrete |
| Fixed shoulder | C70/85 reinforced concrete |
| Central road pavement | Bituminous conglomerate |
| Lateral road pavement | Bankirai wood |

are presented. The calibration of the model was carried out by minimizing the difference between the main experimentally identified eigenfrequencies ($f_{n,\mathrm{EXP}}$) and related mode shapes and the corresponding numerical eigenfrequencies ($f_{n,\mathrm{NUM}}$) and related mode shapes, obtained by changing link element mechanical characteristics in numerical modal analyses performed using SAP2000. Based on the criteria of Eq. 3, considering that tests were performed in ambient vibration condition (microtremors), the configuration that best suits the experimental outcome was the one that included fixed base foundation constraints and linear links for modelling the sliding support devices. The maximum acceptable error ($\varepsilon$) was fixed to 5%.

$$E\left(f_{\mathrm{NUM}},f_{\mathrm{EXP}}\right)=\frac{1}{n}\cdot\sqrt{\sum_{n=1}^{\infty}\left(f_{n,\mathrm{NUM}}-f_{n,\mathrm{EXP}}\right)^{2}}<\varepsilon. \tag{3}$$

Table 4 compares the experimental parameters reported in Table 2 and the value relating to the optimal calibrated model in terms of the main natural frequencies. A good agreement among numerical and experimental eigenfrequencies can be observed.

Using Eq. 3, considering the three detected modes of vibration, the error evaluated for the calibrated model is equal to 2%. Then, the calibrated numerical model is accepted and used to perform the thermo-mechanical analyses.

## 6 Results of thermal analyses

The calibrated numerical model was utilized for thermal analysis of the bridge, aimed at evaluating the elastic deformations of the bridge resulting from temperature fluctuations. The thermal loads were determined based on the temperature recordings presented in Sect. 4.2 and uniformly applied to the structural elements in accordance with the Italian code [39]. The uniform application of thermal loads is representative of the air temperature variations (excluding the effects of the solar radiation). The mathematical constitutive equations used within the numerical analyses are those implemented in SAP2000 software [38].

Several numerical investigations were performed using the calibrated numerical model by changing the value of the temperature difference (from – 15 °C to + 15 °C), thereby obtaining the corresponding values of the maximum deformation in the centreline of the bridge deck. Nonlinearities associated to stress variation of steel tendon have been considered within the numerical analyses. Figure 15 illustrates the maximum vertical displacements evaluated using three different temperature gradients. Specifically, for a temperature variation of – 15 °C, the corresponding maximum displacement is equal to – 5.85 cm, while for temperature variations of 5 °C and 15 °C, the maximum vertical displacement is equal to 2 cm and 5.85 cm, respectively.

The outcomes of the thermal analyses are summarized in Fig. 16. The diagram shows an empirical distribution of the maximum deformations induced by the relative temperature variations.

This distribution was used to establish a correlation (Eq. 4), valid for the monitored bridge, between the variations in air temperature ($\Delta T$), expressed in °C, and the maximum deformations ($f_{\max}$), expressed in m, generated on the bridge deck (Fig. 15).

$$f_{\max}(m)=0.0009+0.004*\Delta T(^{\circ}C). \tag{4}$$

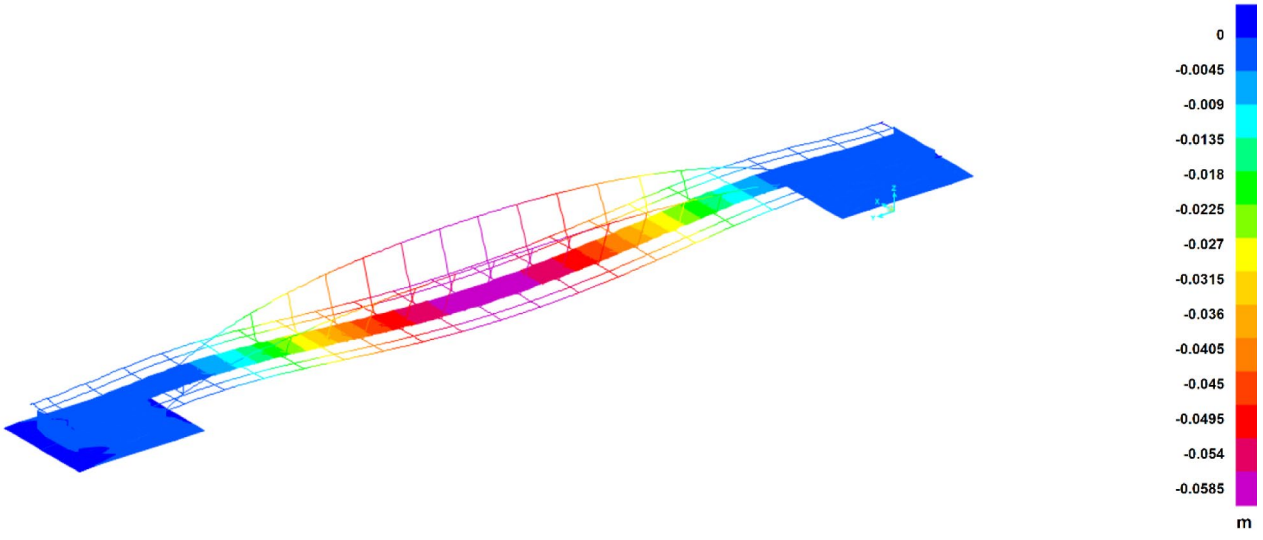
Starting from the available temperature dataset, the empirical cumulative distribution functions (ECDF) of the temperature variation were evaluated to find the most probable temperature variation in the considered period. ECDF is a statistical function showing how often a measured variable has a certain value or less in a sample. The ECDF is a step function that exhibits a jump of 1/n at each of the n data points. The median value of the data distribution corresponds to ECDF = 0.5. Based on the satellite acquisition times and the temperature histories acquired by Boncompagni weather station, the most probable temperature variation between two consecutive acquisitions is ± 5 °C, corresponding to a maximum vertical displacement of ± 2 cm for the bridge deck (Fig. 17).

In Fig. 18, the ascending (blue) and descending (red) orbits of persistent scatterers generated by satellite data processing are integrated with the vertical displacement field of the bridge resulting from the numerical model obtained by considering a temperature variation of – 5 °C.

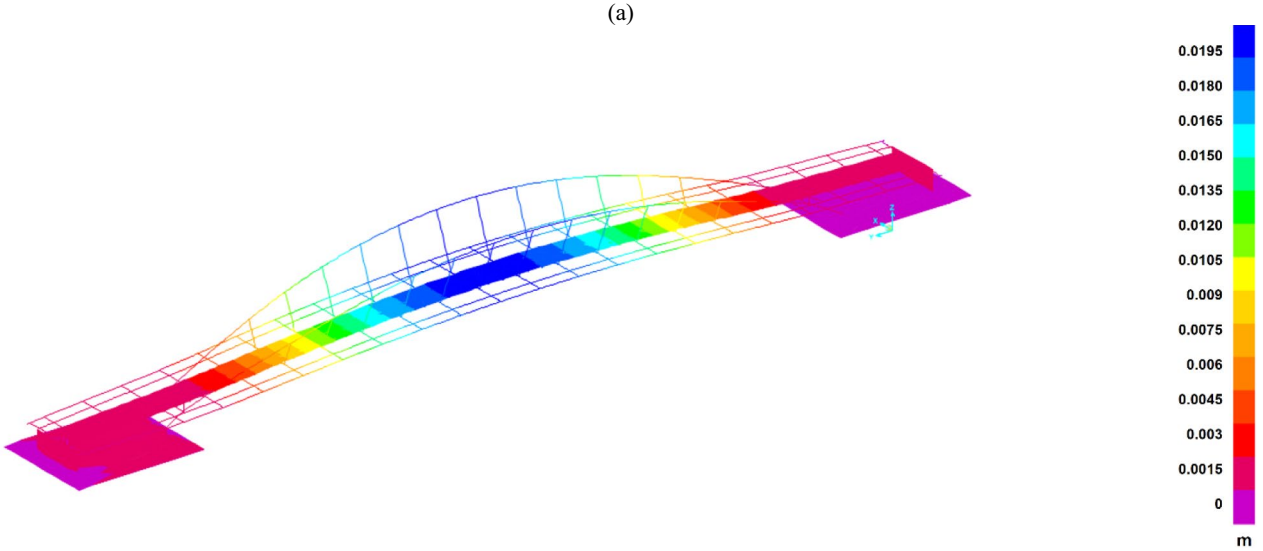
It is noteworthy that the persistent scatterers are exclusively present on the external spans of the bridge, whereas in the central area of the deck, which is characterized by

**Table 4** Main experimental and numerical eigenfrequencies of the bridge

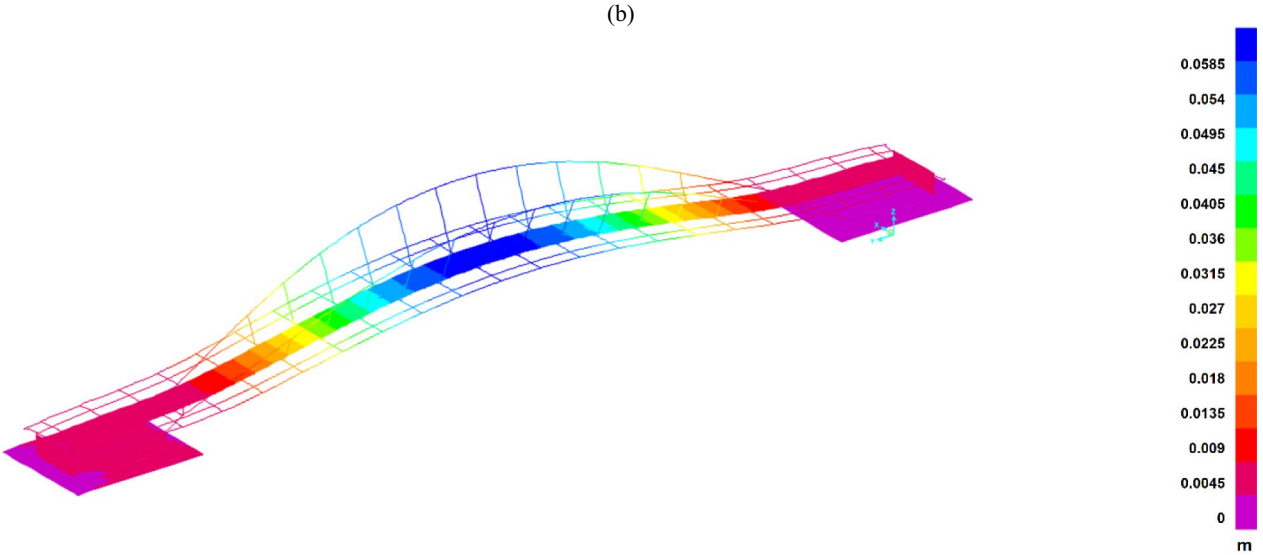
| Mode number | Experimental Frequency [Hz] | Optimal numerical model Frequency [Hz] | Percentage difference Rate [%] |
|---|---|---|---|
| Mode 1 | 0.92 ± 0.01 | 0.92 | 0 ± 0.1 |
| Mode 2 | 1.01 ± 0.01 | 1.07 | 5.94 ± 0.1 |
| Mode 3 | 1.33 ± 0.01 | 1.33 | 0 ± 0.1 |




(a)




(b)




(c)

**Fig. 15** Bridge deformations produced by temperature variations equal to **a** – 15 °C, **b** 5 °C and **c** 15 °C, respectively (note the differences in the scale range)

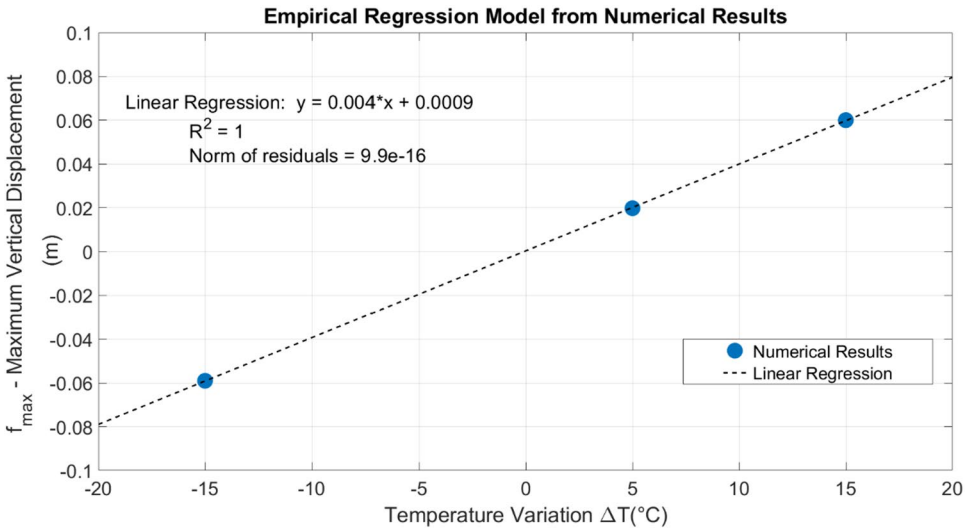



**Fig. 16** Linear regression model between maximum vertical displacement and temperature variation obtained from numerical analyses

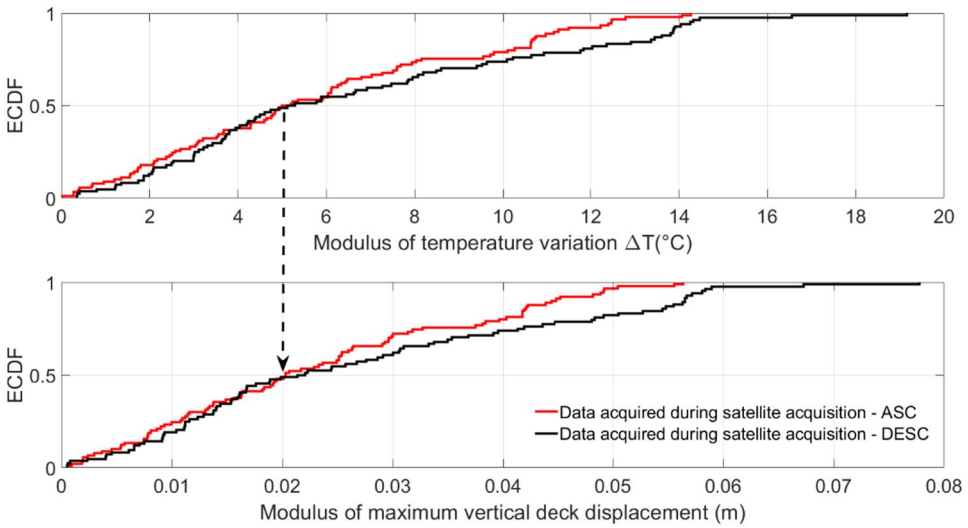



**Fig. 17** Empirical cumulative distribution functions of temperature variations (top) and the associated maximum vertical displacement of the centreline of the deck (bottom)

higher displacement values, they are absent. Equations 1 and 2 provide a relationship that correlates the maximum measurable displacement along the LOS to the wavelength of the sensor installed on the satellite. Based on Eq. 2, the maximum measurable displacement value is approximately 1/4 of the total wavelength. Therefore, for radar systems such as COSMO-SkyMed, working in the X band, characterized by a wavelength of 3.1 cm, the maximum detectable value along the LOS is approximately 0.8 cm. Hence, the displacement values that statistically exceed that value would be ambiguous and, consequently, unquantifiable in the absence of additional information. In these cases, it might be appropriate to integrate the information retrieved from measurements carried out with multiple satellite radar systems operating with lower frequencies, therefore characterized by significantly longer wavelengths, as in the case of sensors operating in L band [40].

However, the linear model considered obtaining the maximum deformations of the deck could return values lower than the real ones because the higher temperature resulting from solar radiation and the possible non-linear behaviour resulting from internal constraints and complex conditions were neglected in the numerical model. In general, the structural response to thermal loads for an infrastructure is influenced not only by air temperature, but also by other components such as solar radiation and structural thermal inertia. The bridge can be viewed as a dynamic system whose response is out of phase with the variation in air temperature, depending on the above-mentioned parameters. Especially during daylight hours, radiation effects could cause an increase in the surface temperature of the bridge relative to that of the air by between 5 and 20 °C.

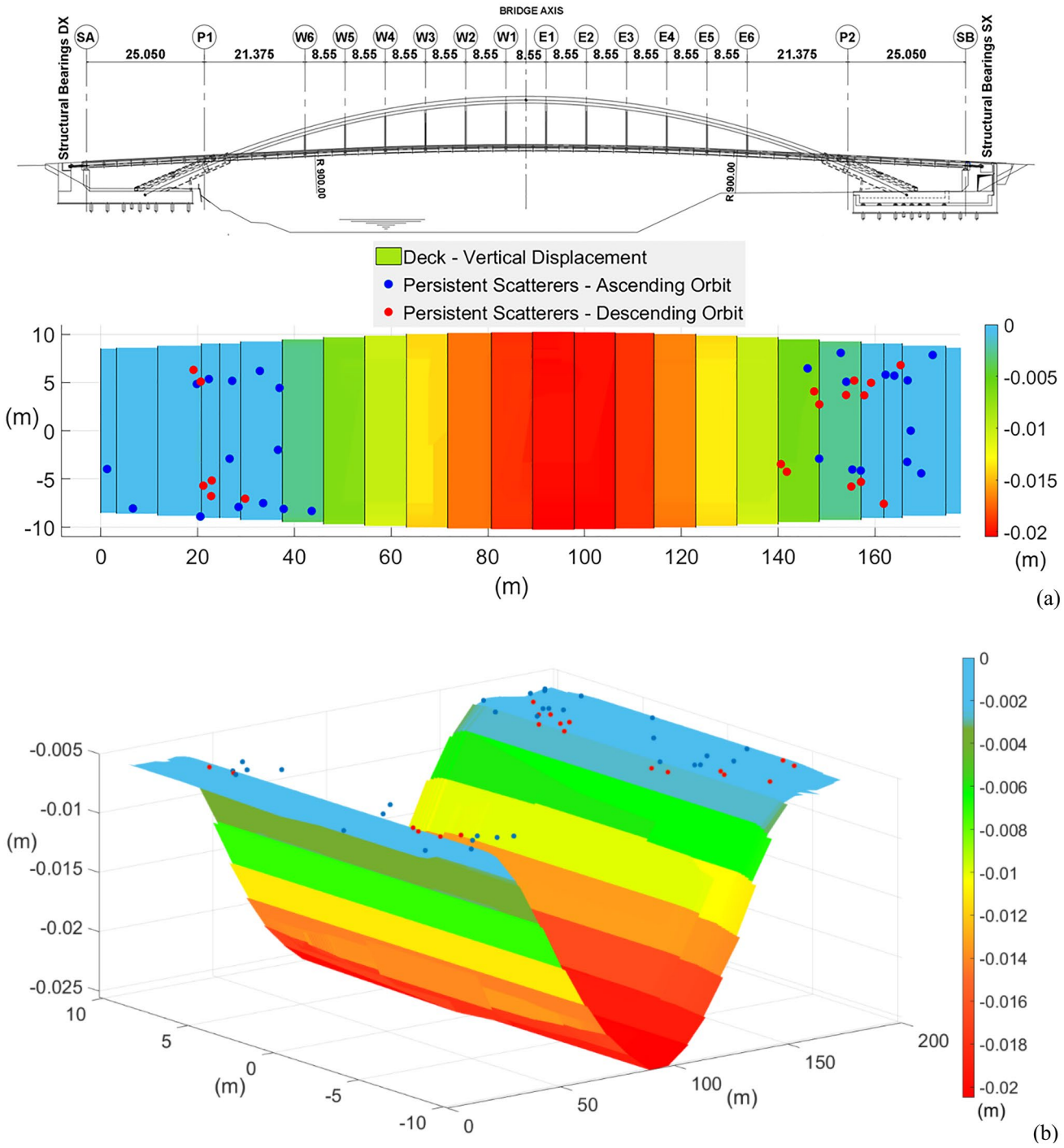



**Fig. 18** Deformed shape of the bridge for – 5 °C of temperature variation

## 7 Discussion and conclusions

One of the main goals of the 2019–2021 and 2022–2024 DPC-ReLUIS Projects—WP6 "Structural Health Monitoring and Satellite Data" was setting up new protocols to merge information retrieved from satellite data and on-site measurements. The "Ponte della Musica–Armando Trovajoli" bridge has been selected as a test site with the aim of understanding the causes that produced the loss of satellite data on the main deck of the bridge. Once retrieved and analysed the design documentation, two in situ experimental campaigns have been planned with the aim of verifying the

structural details and performing vibrational acquisition on the bridge using both velocimetric and accelerometric sensors. Experimental vibrational data were used to evaluate the main dynamic characteristics of the bridge, which were then used to calibrate the related numerical model. The calibration has been performed by acting on the mechanical characteristics of the external constrains, minimizing the difference between numerical and experimental eigenfrequencies and using the equivalent viscous damping factors experimentally evaluated.

Once the numerical model of the bridge is calibrated, the temperature time history recorded at Boncompagni weather station from 2013 to 2019 was analysed to define the air temperature variations to be used as input within the numerical model. Then, considering the real satellite revisit time of COSMO-SkyMed constellation, the ECDF of the temperature variations was evaluated. The statistical parameter used to define the reference temperature variation was the median value that corresponds to the probability equal to 50% in the ECDF, and it is equal to $\pm 5$ °C. Based on the numerical simulations, a temperature variation of $\pm 5$ °C can produce a maximum vertical displacement of the deck of the monitored bridge equal to $\pm 2$ cm with a daily frequency of occurrence. It is important to note that the real values of vertical displacement of the monitored bridge could be higher than the estimates because of the surface temperature increase in the structural elements due to solar radiation. Further analyses are required to generalize the results on other bridges having similar characteristics.

As discussed in previous sections, satellites of the COSMO-SkyMed constellation are equipped with sensors working within the X band, characterized by a wavelength of 3.1 cm [41]. Considering Eq. 2, the maximum displacement recordable along the LOS direction is a fraction of the total wavelength and, as discussed in Sect. 6, his value is less than 1 cm. Therefore, the displacement values that statistically exceed that value would be ambiguous and, consequently, unquantifiable in the absence of additional information. In such cases, it may be advantageous to integrate data obtained from measurements conducted with multiple satellite radar systems operating at lower frequencies, thus working on significantly longer wavelengths, as observed in the case of sensors operating in the L band. Alternatively, it could be possible to integrate satellite data with on-site information systems such as global navigation satellite system (GNSS), terrestrial interferometric synthetic aperture radar (TInSAR), optical fibre strain monitoring systems and light detection and ranging (LiDAR).

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**Author contributions** FCP and RD: conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing—original draft, funding, writing—review and editing, supervision; FCP, RD, GA: experimental data acquisitions; FCP, RD, GA, PI: numerical modelling. All authors read and approved the final manuscript.

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**Data availability** The authors state that vibrational acquisition data will be available on request.

## Declarations

**Conflict of interest** No relevant financial or non-financial interests have been disclosed by the authors. The authors do not have any competing interests to declare that are pertinent to the content of this article.

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